USN | | | | | | | | | | [22ESEC31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

COMPUTER ORGANIZATION AND ARCHITECTURE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer All Questions

PART-A **5 X 2=10**

1.	**a.**	Convert $(101010011.110100)_{10} = (\ ?\ )_8$	L2	CO1	**02**
	b.	Sketch the basic functional units of a computer.	L1	CO1	**02**
	c.	List the phases of instruction cycle.	L1	CO2	**02**
	d.	Differentiate between SRAM and DRAM.	L2	CO3	**02**
	e.	List any two logical instructions and explain them with an example.	L2	CO4	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Distinguish between RISC and CISC.	L2	CO1	**05**
		OR			
	b.	Illustrate the connection between processor and memory with its operational steps.	L2	CO1	**05**
3.	**a.**	Perform the 2's complement subtraction of smaller number (101011) from larger number (111001)	L3	CO1	**05**
		OR			
	b.	Perform arithmetic operation in binary numbers with negative numbers in signed 2's Complement form (-35) + (-40).	L3	CO1	**05**
4.	**a.**	Multiply the two numbers 7 and 3 by using the Booth's multiplication algorithm.	L2	CO2	**05**
		OR			
	b.	List and explain the various Registers of a Basic Computer.	L2	CO2	**05**
5.	**a.**	Discuss the Salient features of USB operation.	L2	CO3	**05**
		OR			
	b.	Define the following: **i.** Addressing modes. **ii.** microinstruction **iii.** microprogram **iv.** microcode.	L2	CO3	**05**
6.	**a.**	Explain in detail about secondary storage devices with neat diagram.	L2	CO4	**05**
		OR			
	b.	Give comparison between memory mapped I/O and I/O mapped I/O.	L2	CO4	**05**

7.	a.	Compare the various mapping used in cache memory.	L3	CO5	05
		OR			
	b.	List the different types of interrupts and explain.	L2	CO5	05
		PART-C	**6 X 10=60**		
8.	a.	Describe how you'll evaluate a computer's performance. What are the criteria used to evaluate a computer's performance? Explain Basic Performance Equation.	L2	CO1	10
		OR			
	b.	With a neat diagram explain the connections between the different processor register and the memory.	L2	CO1	10
9.	a.	Explain the different types of addressing Modes.	L3	CO2	10
		OR			
	b.	Demonstrate how the following expression X = (A + B) * (C + D) will be executed in one address, two address, three address-based processors.	L3	CO2	10
10.	a.	Define Micro-operation. Design the Hardware implementation for Shift Micro operations and also explain with suitable example.	L3	CO2	10
		OR			
	b.	Consider a two level with access times of 5 ns, and 80 ns respectively. If the hit rates are 92%, and 75% respectively in the two caches and the memory access time is 250 ns, what is the average access time?	L4	CO2	10
11.	a.	Describe the address sequencing capabilities in control memory with neat sketch.	L3	CO3	10
		OR			
	b.	Demonstrate the organization of micro – programmed control unit with block diagram.	L3	CO3	10
12.	a.	Discuss the generation of Athlon processor.	L3	CO4	10
		OR			
	b.	Explain in detail about secondary storage devices with neat diagram.	L3	CO4	10
13.	a.	Write a short note on instruction cycle in Pentium 4 processor.	L3	CO5	10
		OR			
	b.	Perform the following on A_t=1100 B=1010 A_{t+1}=? **i.** Selective set **ii.** Selective complement **iii.** Selective clear **iv.** Mask	L3	CO5	10